

MITHUN LAKSHMIPATHY

| slpmithun@gmail.com | +91 87544 87755 | linkedin.com/in/mithunlakshmipathy

| github.com/mithun-28 | https://huggingface.co/mithun-ai |

PROFILE

AI/ML Engineer (2026 graduate) with hands-on experience building production RAG systems, conversational AI, and predictive ML models in Python, LangChain, and Scikit-learn. Delivered AI-driven product features at Zuntra Digital across 7 months; achieved 92% accuracy on a deployed healthcare prediction model. Looking to join a product-focused startup as an ML Engineer and own the full build-to-deploy lifecycle.

SKILLS

Programming Languages	: Python, SQL, R
AI and Machine Learning	: PyTorch, Scikit-learn, HuggingFace Transformers, BERT, OpenAI API, LangChain, RAG, NLP, Prompt Engineering
MLOps and Deployment	: Docker, FastAPI, Django, REST API Development, Git, Model Evaluation
Database and Backends	: SQL, NoSQL, Firebase, FAISS, API Integration
Data Analysis and Visualization	: Pandas, NumPy, Matplotlib, Power BI, MS Excel, Google Sheets
Soft Skills	: Analytical Thinking, Problem Solving, Stakeholder Collaboration, Data-Driven Decision Making, Product-Oriented Thinking

WORK EXPERIENCE

Contus Tech | *Artificial Intelligence Intern*

Porur, Chennai | May 2026 – June 2026

- Developed AI application components using LangChain and FastAPI, integrating SQL and NoSQL datastores for a live product environment.
- Containerised AI services using Docker, streamlining deployment consistency across development and staging environments.
- Contributed to cross-functional product team workflows, participating in sprint planning and AI feature scoping for production-grade systems.

Zuntra Digital | *AI Product Solutions Associate*

Perungudi, Chennai | Sep 2025 – Apr 2026

- Built Python-based data analysis pipelines to surface product insights, directly informing feature prioritisation decisions across multiple product cycles.
- Designed and delivered Power BI dashboards tracking key product KPIs, enabling stakeholders to make faster, data-backed decisions.
- Managed end-to-end SQL database workflows including schema design, data quality checks, and pipeline maintenance supporting live analytics systems.

QUAZR Motors | *Intern - Data Engineering and Analytics*

Siruseri, Chennai | Aug 2023 – Dec 2023

- Built and validated ML classification models using Python and Scikit-learn to power driver profiling features, improving prediction accuracy by 18%.
- Engineered end-to-end real-time sensor data pipelines covering ingestion, preprocessing, and feature engineering, enabling reliable downstream model performance.
- Designed SQL schemas and implemented data quality checks supporting scalable analytics pipelines across live vehicle telemetry data.

PROJECTS

DataTalk: Hybrid CASA – Conversational NL-to-SQL Analytics System

- Built a schema-aware NL-to-SQL platform using Python, HuggingFace Transformers, and REST APIs, enabling non-technical users to query enterprise databases in plain English.
- Implemented context-aware query generation with transformer-based semantic parsing, achieving accurate SQL outputs across complex multi-table schemas.
- Eliminated SQL dependency for business users, cutting query turnaround from hours to seconds for analytical decision-making workflows.

Predictive Analytics for Hypertension Risk

- Trained and evaluated Scikit-learn classification models with SHAP-based interpretability and AutoML, achieving 92% prediction accuracy on clinical hypertension data.
- Dockerised the trained model and deployed a REST API endpoint with a Django-based prediction GUI — full build-to-deploy pipeline owned end-to-end.
- Integrated SHAP explainability layer to surface feature-level risk drivers, making the model output actionable for non-technical healthcare stakeholders.

Domain-Specific RAG based Question Answering Chatbot System

- Built a retrieval-augmented generation pipeline using FAISS vector store, HuggingFace Transformers, and custom prompt engineering over a 10k+ document corpus.
- Designed the end-to-end retrieval architecture — chunking strategy, embedding generation, similarity search, and context-aware LLM response synthesis.
- Achieved sub-2s response latency with high answer relevance, significantly reducing manual document lookup dependency.

Meta-Reinforcement Learning Cloud Scheduler

- Built a PPO-based Meta-RL scheduler in Python to dynamically allocate cloud compute resources based on real-time workload patterns.
- Integrated explainability-driven monitoring to surface scheduling decisions, improving load balancing transparency and operational trust.
- Optimised resource utilisation across variable workload distributions, demonstrating applied RL beyond standard supervised ML use cases.

ACHIEVEMENTS & ACKNOWLEDGEMENTS

Voyage 2024 Hackathon - Top 20 finalist (100 teams) | Built an end-to-end AI finance product featuring intelligent tax planning, savings optimisation, and personalised financial insights using Python and LLM-based reasoning.

EDUCATION

Vellore Institute of Technology

M Tech Integrated CSE with specialization in Business Analytics

Chennai, Tamil Nadu | 2021 - 2026

CGPA: 7.87/10.00

VOLUNTEERING

National Service Scheme (NSS) | Volunteer

VIT Chennai | April 2023 – June 2023

- Created social media content, organized awareness drives, and educated rural students, boosting outreach and team-based project coordination.